

02/04/2026

Toric geometry VI:

T-equivariant
divisors & polytopes

Last time Properness, polytopes, recollections on divisors

Today T-equivariant Cartier divisors & polytopes

Correction from last time. Given a rational convex polytope $P \subset M_{\mathbb{R}}$, for each face $Q \subset P$, we defined

$$\sigma_Q^\vee := \text{Cone} \{u' - u \mid u' \in P, u \in Q \text{ vertices}\},$$

and $\sigma_Q = (\sigma_Q^\vee)^\vee$. We said that the normal fan assoc to P was

$$\{\sigma_Q \mid Q \subset P \text{ proper face}\}$$

We should have said

$$\Sigma_P = \{\sigma_Q \mid Q \subset P \text{ nonempty face}\}$$

Then $\sigma_P^\vee = 0$, so $\sigma_P = 0$.

> This has been corrected in the notes

Linear Equivalence

Q When should we regard two divisors as the same?

A If $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$ Equivalently, if

$$\mathcal{O}_X(D - D') = \mathcal{O}_X(D) \otimes \mathcal{O}_X(-D')$$

is isomorphic to $\mathcal{O}_X$. That is, $[\mathcal{O}_X(D)] = [\mathcal{O}_X(D')] in $\text{Pic}(X)$.$

Q When is $\mathcal{O}_X(-D) \cong \mathcal{O}_X$?

A If there is a nowhere vanishing global section $f: \mathcal{O}_X \rightarrow \mathcal{O}_X(-D)$.
Equivalently, if D is cut out by a single global section.

Def An effective Cartier divisor D is principal if there is a global section $f \in \Gamma(X, \mathcal{O}_X)$ such that $\langle f \rangle = \mathcal{O}_X(-D)$.

> So D is the vanishing locus of f .
↑ ideal sheaf generated by f

Then $\text{Div}^+(X) = \{\text{effective Cartier divisors on } X\}$

> Then $\text{Div}^+(X)$ is a commutative monoid under $+$ with 0 the empty divisor

> We have a map

$$\begin{array}{ccc} D & \xrightarrow{\quad} & \mathcal{O}_X(D) \\ \text{Div}^+(X) & \xrightarrow{\quad} & \text{Pic}(X) \\ \downarrow & \nearrow \exists! & \\ \text{Div}^+(X)^{\text{gp}} & \xleftarrow{\text{group completion}} & \end{array}$$

Fact. If M is a cancellative monoid (i.e., $xy = xz \Rightarrow y = z$), then the natural map $M \rightarrow M^{\text{gp}}$ is injective.

Ex. $\text{Div}^+(X)$ is cancellative.

Problem. $U \mapsto \text{Div}^+(U)$ is a sheaf, but $U \mapsto \text{Div}^+(U)^{\text{gp}}$ isn't.

Let X be a scheme. Write $\underline{\text{Div}}^+ = \underline{\text{Div}}_X^+$ for the sheaf of comm monoids

$$U \mapsto \underline{\text{Div}}_X^+(U)$$

> Write $\underline{\text{Div}} = \underline{\text{Div}}_X$ for the sheafification of the presheaf

$$U \mapsto \underline{\text{Div}}_X^+(U)^{\text{gp}}$$

 In general, $\underline{\text{Div}}_X^+(U)^{\text{gp}} \rightarrow \underline{\text{Div}}(U)$ is not an isomorphism

Def. The abelian group of Cartier divisors on X is

$$\text{Div}(X) := \underline{\text{Div}}(X)$$

> The Cartier class group of X is

$$\text{DivCl}(X) := \text{Div}(X) / \text{principal divisors}$$

Theorem Let X be a scheme

(1) If X is integral, there is a short exact sequence

$$1 \longrightarrow \Gamma(X, \mathcal{O}_X) \longrightarrow K(X)^\times \longrightarrow \text{Div}(X) \longrightarrow \text{DivCl}(X) \longrightarrow 1$$

(2) If X is integral or is locally noetherian with an open dense affine subscheme, then the homomorphism

$$\text{Div}(X) \longrightarrow \text{Pic}(X)$$

induced by $D \mapsto \mathcal{O}_X(D)$ is surjective and induces an isomorphism

$$\text{DivCl}(X) \xrightarrow{\sim} \text{Pic}(X)$$

Weil divisors

needed so dimension theory is well-behaved

Recall. Let X be a noeth. scheme. The group of Weil divisors on X is

$$\mathbb{Z}^1(X) = \mathbb{Z} \left\langle \begin{array}{l} \text{closed integral subschemes} \\ C \subset X \text{ of codim } 1 \end{array} \right\rangle$$

There is always a group homomorphism

$$\text{cyc}: \text{Div}(X) \longrightarrow \mathbb{Z}^1(X)$$

$$D \longmapsto \sum_C \text{ord}_C(D) [C]$$

└ vanishing order

- > If X is normal, then cyc is injective.
- > If X is locally factorial (all local rings are UFDs), then cyc is an isomorphism.

Exercise Let A be an integrally closed domain and $\sigma \subset N_{\mathbb{R}}$ a cone. Then $A\langle S_{\sigma} \rangle$ is an integrally closed domain.

> Hence if Σ is a fan in N , then $X_{\Sigma, A}$ is a normal scheme.

Convention For the rest of the lecture, we'll work over a field k , which we leave implicit (so X_{Σ} means $X_{\Sigma, k}$). This is because we want to make sure that the group of Cartier divisors on the base ring is trivial.

> This isn't true for $\mathbb{Z}$. $\text{Div}(\mathbb{Z}) \simeq \mathbb{Q}^{\times} / \{\pm 1\}$.

Definition (ad hoc, but can be improved) Let (N, Σ) be a fan. The group of T -equivariant weil divisors on X_{Σ} is

$$\mathbb{Z}_T^1(X_{\Sigma}) = \mathbb{Z} \left\langle \begin{array}{l} T\text{-equivariant integral} \\ \text{closed codim } 1 \text{ } C \subset X \end{array} \right\rangle$$

The group of T -equivariant Cartier divisors on X_{Σ} is

$$\text{Div}_T(X_{\Sigma}) := \text{Div}(X_{\Sigma}) \cap \mathbb{Z}_T^1(X_{\Sigma}).$$

Orbit Closures

Goal Understand some T -equivariant principal Weil divisors

> The whole story we'll tell is the obvious generalization of trying to understand the orbits of the action of $(\mathbb{C}^*)^n$ on $\mathbb{C}^n$

Notation. Let (N, Σ) be a fan and $\tau \in \Sigma$. Write

$$N_\tau := \tau \cap N + (-\tau \cap N)$$

be the sublattice generated by $\tau \cap N$, and

$$N(\tau) := N/N_\tau, \quad M(\tau) := \tau^\perp \cap N$$

Then $N(\tau)$ and $M(\tau)$ are dual lattices

Def. The star of τ is the fan in $N(\tau)$ consisting of the cones

$$\bar{\sigma} := \text{image of } \sigma \text{ under } N \rightarrow N(\tau)$$

for $\sigma \supset \tau$ a cone in Σ . We denote this fan by $\text{star}(\tau)$.

> As a poset, $\text{star}(z) \simeq \{\sigma \in \Sigma \mid \sigma > z\}$

> We write $V_z := X_{\text{star}(z)}$

> Note that V_z is tonic of dimension $\text{rank}(N) - \dim(z)$

- In particular, if z is an edge, V_z has dim 1 less than X_Σ .

Construction (embedding $V_z \hookrightarrow X_\Sigma$). For each $\sigma > z$ note that we have a surjective ring hom

$$\mathbb{Z}\langle \sigma^\vee \cap M \rangle \longrightarrow \mathbb{Z}\langle \underbrace{\sigma^\vee \cap z^\perp \cap M}_{= \bar{\sigma}^\vee \cap M(z)} \rangle$$

given on generators $u \in \sigma^\vee \cap M$ by

$$u \longmapsto \begin{cases} u, & u \in \sigma^\vee \cap z^\perp \cap M \\ 0, & \text{else.} \end{cases}$$

⌊ zero in the ring... this isn't ind. by a lattice hom.

> On spectra, this induces a closed immersion

$$\begin{array}{ccc} X_{\sigma} & \hookrightarrow & X_{\sigma} \\ \cap & & \cap \\ V_{\tau} & & X_{\Sigma} \end{array}$$

> It is immediate from the definitions that these maps are compatible, and glue to a closed immersion

$$V_{\tau} = \operatorname{Colim}_{\substack{\sigma \in \Sigma \\ \sigma \supset \tau}} X_{\sigma} \hookrightarrow \operatorname{Colim}_{\sigma \in \Sigma} X_{\sigma} = X_{\Sigma}$$

Note If $\tau' \subset \tau$ is a face, we get a closed immersion

$$V_{\tau} \longrightarrow V_{\tau'}$$

> $\tau \mapsto V_{\tau}$ defines a functor

$$\Sigma^{\text{op}} \longrightarrow \left\{ \begin{array}{l} \text{closed subschemes} \\ \text{of } X_{\Sigma} \end{array} \right\}$$

Observe The closed subscheme $V_\Sigma \subset X_\Sigma$ is T -stable

> So if $\rho \in \Sigma$ is a ray (edge), then V_ρ is a T -equivariant Weil divisor.

- We denote this divisor by D_ρ and let $v_\rho \in \rho \cap N$ be the smallest integral vector on the ray.

T-equivariant Cartier divisors

Notation Let N be a lattice and $\sigma \in N_{\mathbb{R}}$ a cone. Write

$$\text{div}: M \longrightarrow \mathbb{Z}_T^1(X_\sigma)$$

for the homomorphism

$$u \longmapsto \sum_{\substack{\rho \in \sigma \\ \text{edge}}} \langle u, v_\rho \rangle D_\rho$$

Proposition. The homomorphism div surjects onto $\text{Div}_T(X_\sigma)$ and has kernel $M(\sigma) = \sigma^\perp \cap M$.

⚠ This does not say every D_ρ is Cartier. Consider our favorite example $\sigma = \text{cone}(2e_1 - e_2, e_2)$. The T-Weil divisors corresponding to $2e_1 - e_2$ and e_2 are straight lines on the cone D_1 and D_2 . If $u = (a, b) \in \mathbb{Z}^2$, then

$$\operatorname{div}(u) = (2a-b)D_1 + bD_2.$$

So D_1 and D_2 aren't in the image of div (but $2D_1$ and $2D_2$ are)

Cor. Let (N, Σ) be a fan. Then

$$\operatorname{Div}_T(X_\Sigma) \simeq \varinjlim_{\sigma \in \Sigma^\circ} M/M(\sigma)$$

Note If we write $\Sigma_{\max} \subset \Sigma$ for the collection of maximal cones, then we deduce that

$$\operatorname{Div}_T(X_\Sigma) \simeq \ker \left(\bigoplus_{\sigma \in \Sigma_{\max}} M/M(\sigma) \longrightarrow \bigoplus_{\sigma, \sigma' \in \Sigma_{\max}} M/M(\sigma \cap \sigma') \right).$$

Proof of Cor.

Since Div_T is a sheaf and $(X_\sigma)_{\sigma \in \Sigma}$ is a cover by T -stable open subschemes that is closed under finite intersection,

$$\text{Div}_T(X) \cong \lim_{\sigma \in \Sigma^{\circ P}} \text{Div}_T(X_\sigma)$$

So the claim is immediate from the previous result. $\square$

Relation to functions

Def Let (N, Σ) be a fan. A continuous function $f: |\Sigma| \rightarrow \mathbb{R}$ is piecewise linear if:

(1) For each cone $\sigma \in \Sigma$, the restriction $f|_{\sigma}$ is linear
 $\uparrow$ contained in a
Subspace, so makes sense

(2) $f(|\Sigma| \cap N) \subset \mathbb{Z}$

Write $C^{PL}(|\Sigma|, \mathbb{R})$ for the collection of piecewise linear functions. Note that PL functions form an abelian group under addition.

Construction Let $D \in \text{Div}_T(X_{\Sigma})$ correspond to compatible elements $(u(\sigma))_{\sigma \in \Sigma} \in \varprojlim_{\sigma \in \Sigma^{\text{op}}} M/N(\sigma)$. Define a function

$$\psi_D: |\Sigma| \longrightarrow \mathbb{R}$$

by

$$\psi_D(v) = \langle u(\sigma), v \rangle \text{ for } v \in \sigma.$$

Note that $M/(\sigma^\perp \cap M)$

is dual to $\sigma \cap N$ so this makes sense

Theorem Let (X, Σ) be a fan. The group homomorphism

$$\text{Div}_T(X_\Sigma) \longrightarrow C^{\text{PL}}(|\Sigma|, \mathbb{R})$$

$$D \longmapsto \psi_D$$

is an isomorphism.

The polyhedron of a divisor

Def. Let (N, Σ) be a fan and $D \in \text{Div}_T(X_\Sigma)$. The polyhedron of D is

$$P_D = \{u \in M_{\mathbb{R}} \mid \langle u, - \rangle \geq \psi_D \text{ on } |\Sigma|\}$$

This is a rational convex polyhedron.

Lemma. There is an isomorphism

$$\Gamma(X_\Sigma, \mathcal{O}(D)) \cong k\langle P_D \cap M \rangle$$

Cor. If (N, Σ) is a proper fan, then $P_D \cap M$ is finite.

Hence P_D is a rational polytope.

Proof.

By the finiteness of coherent cohomology,

$\dim_k \Gamma(X_\Sigma, \mathcal{O}(D))$ is finite.

□

Construction. Let's go the other way. Let $P \subset M_{\mathbb{R}}$ be a convex rational polytope of dimension $\text{rank}(N)$, and consider the toric scheme X_P associated to the normal fan Σ_P .

> Then there is a natural T -equivariant Cartier divisor D_P on X_P corresponding to the PL function

$$\begin{aligned}\psi_P(v) &:= \min_{u \in P} \langle u, v \rangle = \min_{u \in P \cap M} \langle u, v \rangle \\ &= \min_{\substack{u \in P \\ \text{vertex}}} \langle u, v \rangle.\end{aligned}$$

Proposition. In this situation:

(1) The line bundle $\mathcal{O}(D_P)$ is ample.

(2) $P_{D_P} = P$.

Theorem. Let N be a lattice. The maps

$$\begin{array}{ccc}
 (\Sigma, D) & \xrightarrow{\quad} & P_D \\
 \left\{ (\Sigma, D) \mid \begin{array}{l} \Sigma \text{ proper fan in } N, \\ D \in \text{Div}_T(X_\Sigma) \text{ ample} \end{array} \right\} & \begin{array}{c} \xrightarrow{\quad} \\ \xleftarrow{\quad} \end{array} & \left\{ P \subset M_{\mathbb{R}} \mid \begin{array}{l} \text{full dim} \\ \text{rational} \\ \text{polytope} \end{array} \right\} \\
 (\Sigma_P, D_P) & \xleftarrow{\quad} & P
 \end{array}$$

are inverse bijections. Moreover, if Σ is a fan in N and X_Σ is projective, then $X_\Sigma \cong X_P$ for some rational polytope $P \subset M_{\mathbb{R}}$.